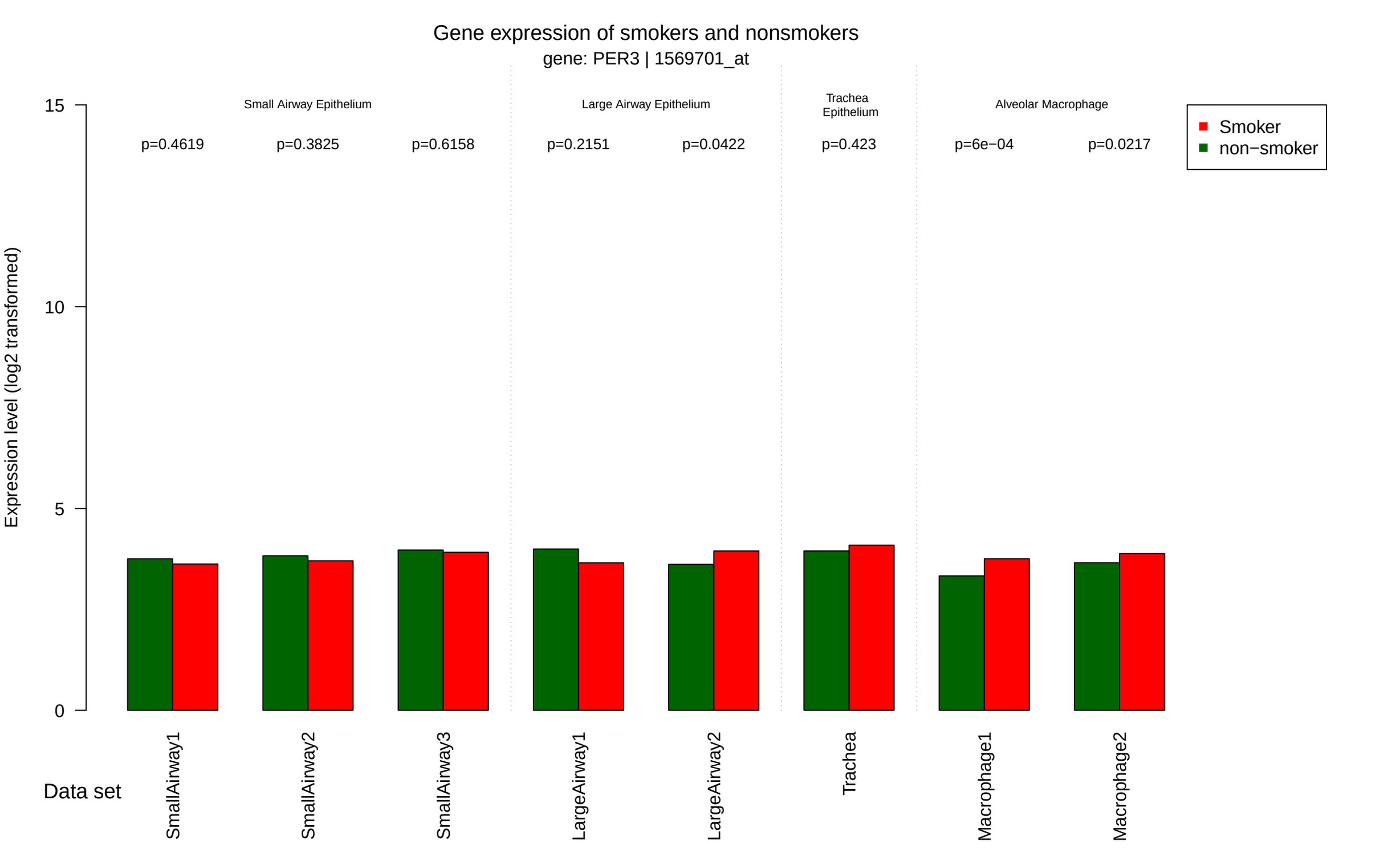
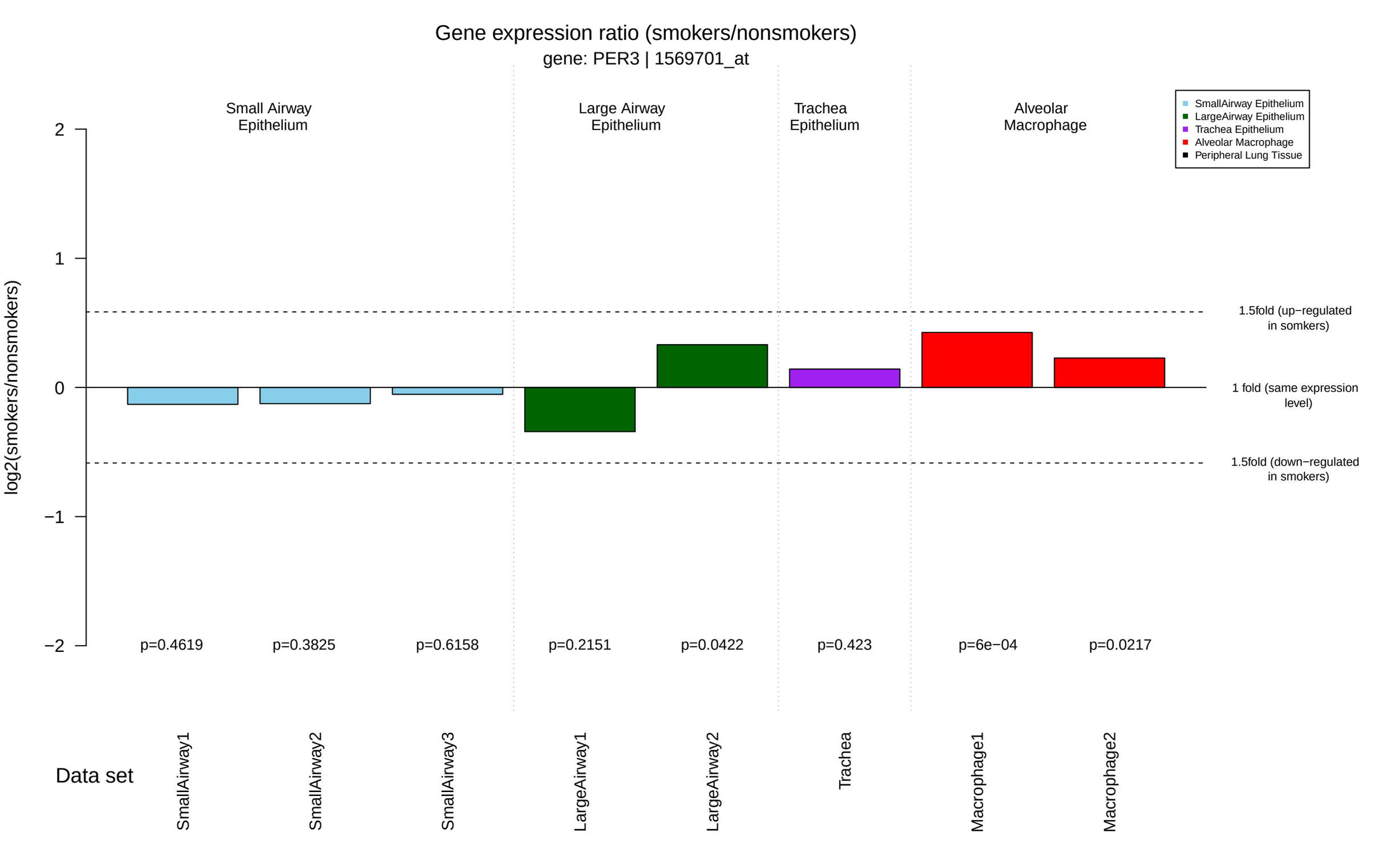
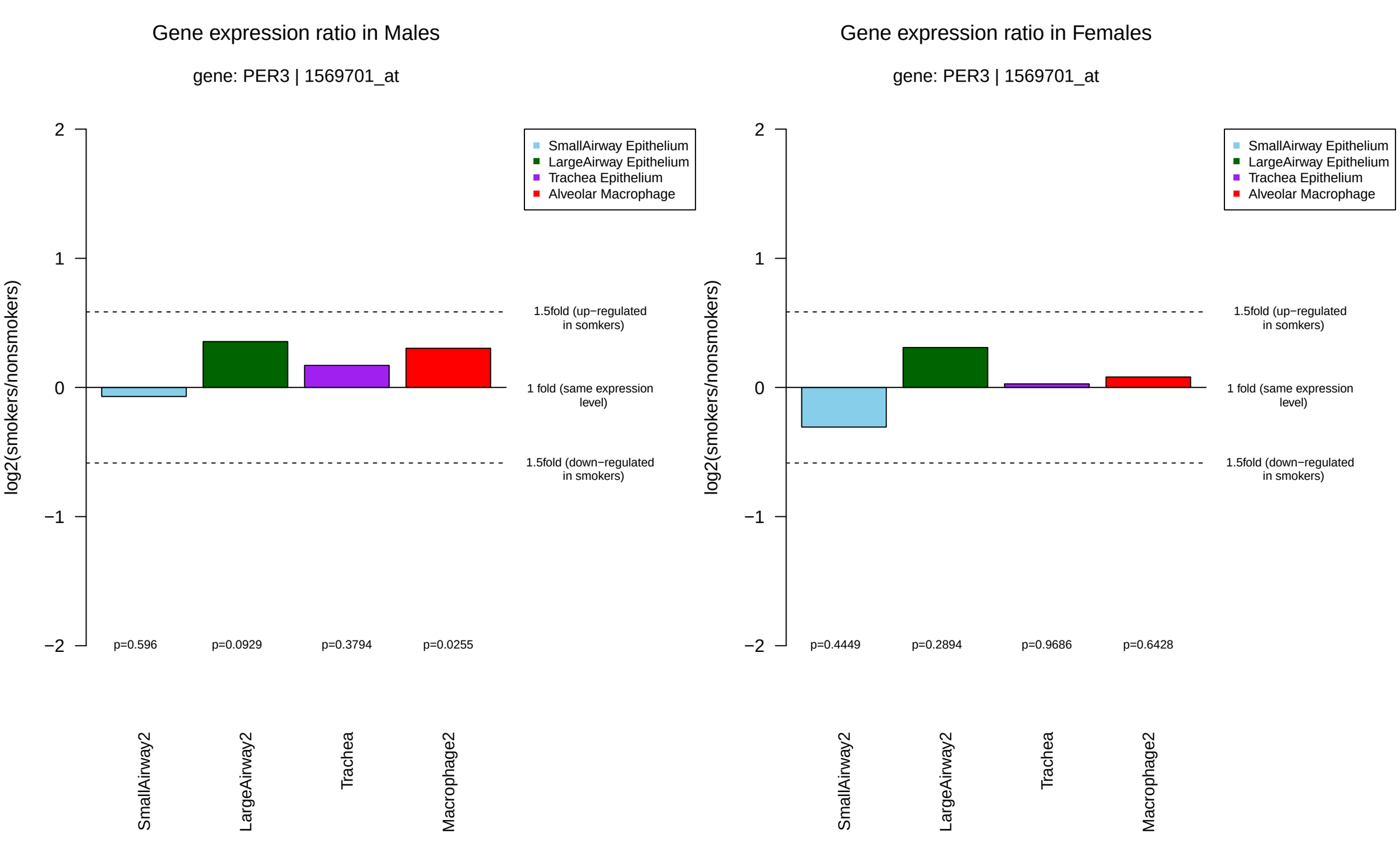
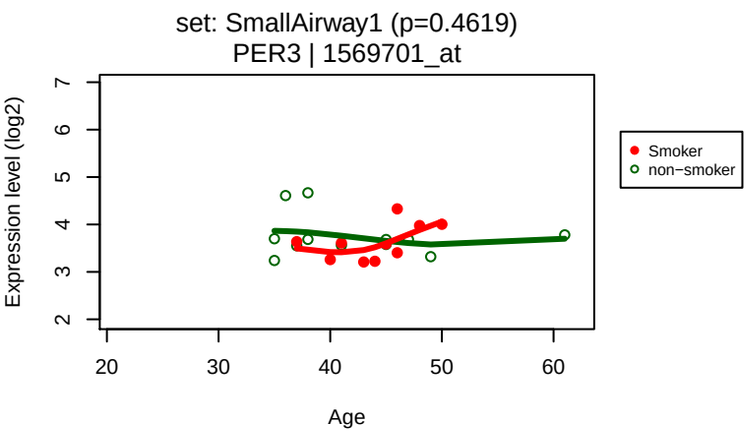
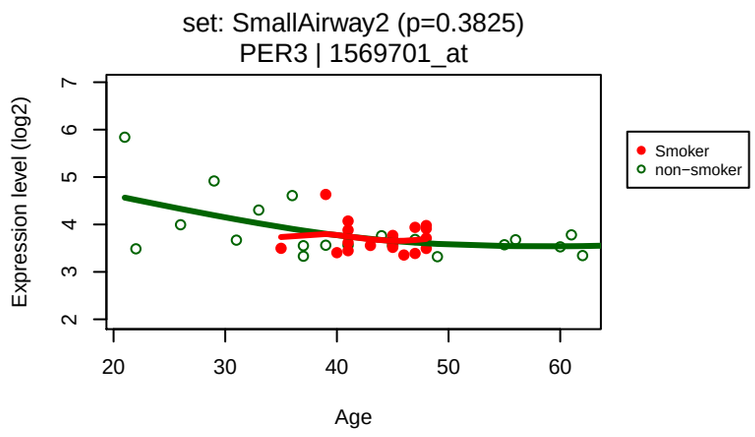
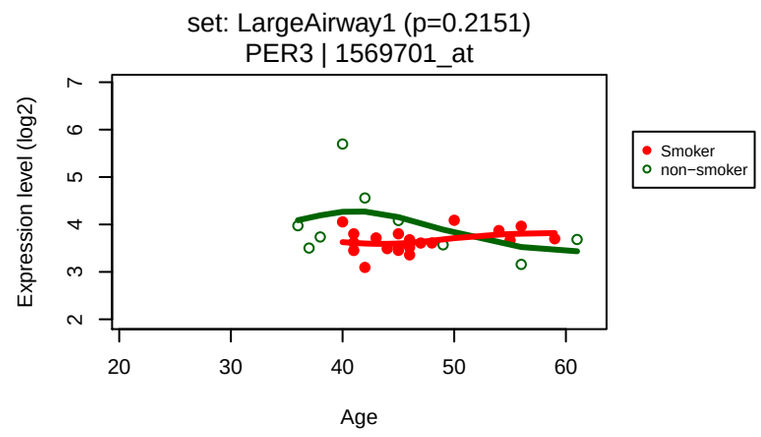

Gene expression of smokers and nonsmokers
gene: PER3 | 1569701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4619
p=0.3825
p=0.6158
p=0.2151
p=0.0422
p=0.423
p=6e−04
p=0.0217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

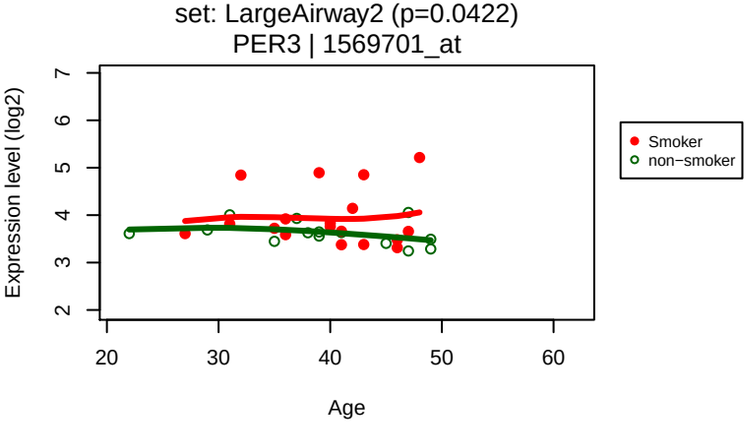

Gene expression ratio (smokers/nonsmokers)
gene: PER3 | 1569701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4619
p=0.3825
p=0.6158
p=0.2151
p=0.0422
p=0.423
p=6e−04
p=0.0217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

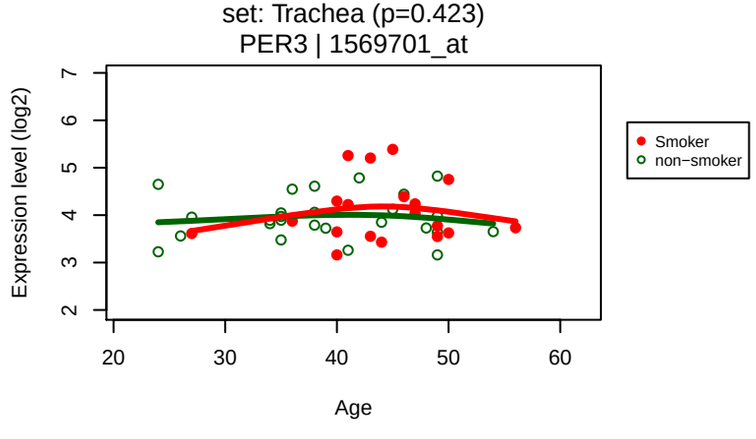

Gene expression ratio in Males
gene: PER3 | 1569701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.596
p=0.0929
p=0.3794
p=0.0255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER3 | 1569701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4449
p=0.2894
p=0.9686
p=0.6428
SmallAirway2
LargeAirway2
Trachea
Macrophage2

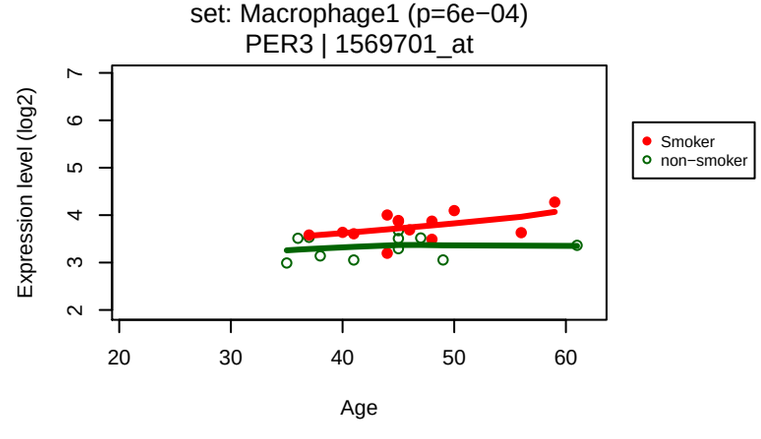

set: SmallAirway1 (p=0.4619)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker

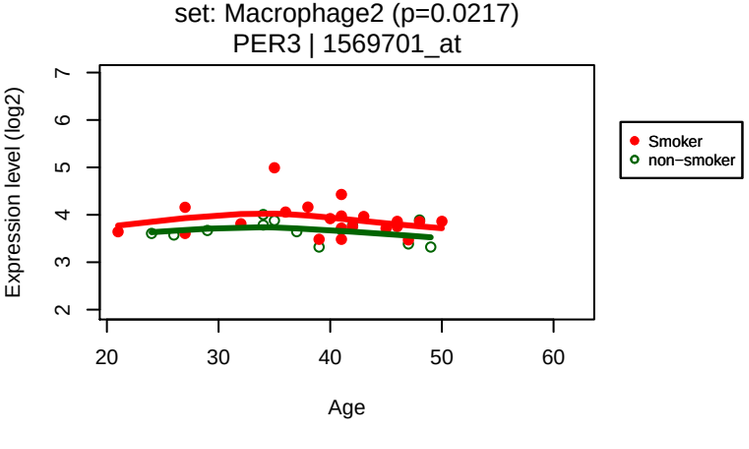

set: SmallAirway2 (p=0.3825)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker

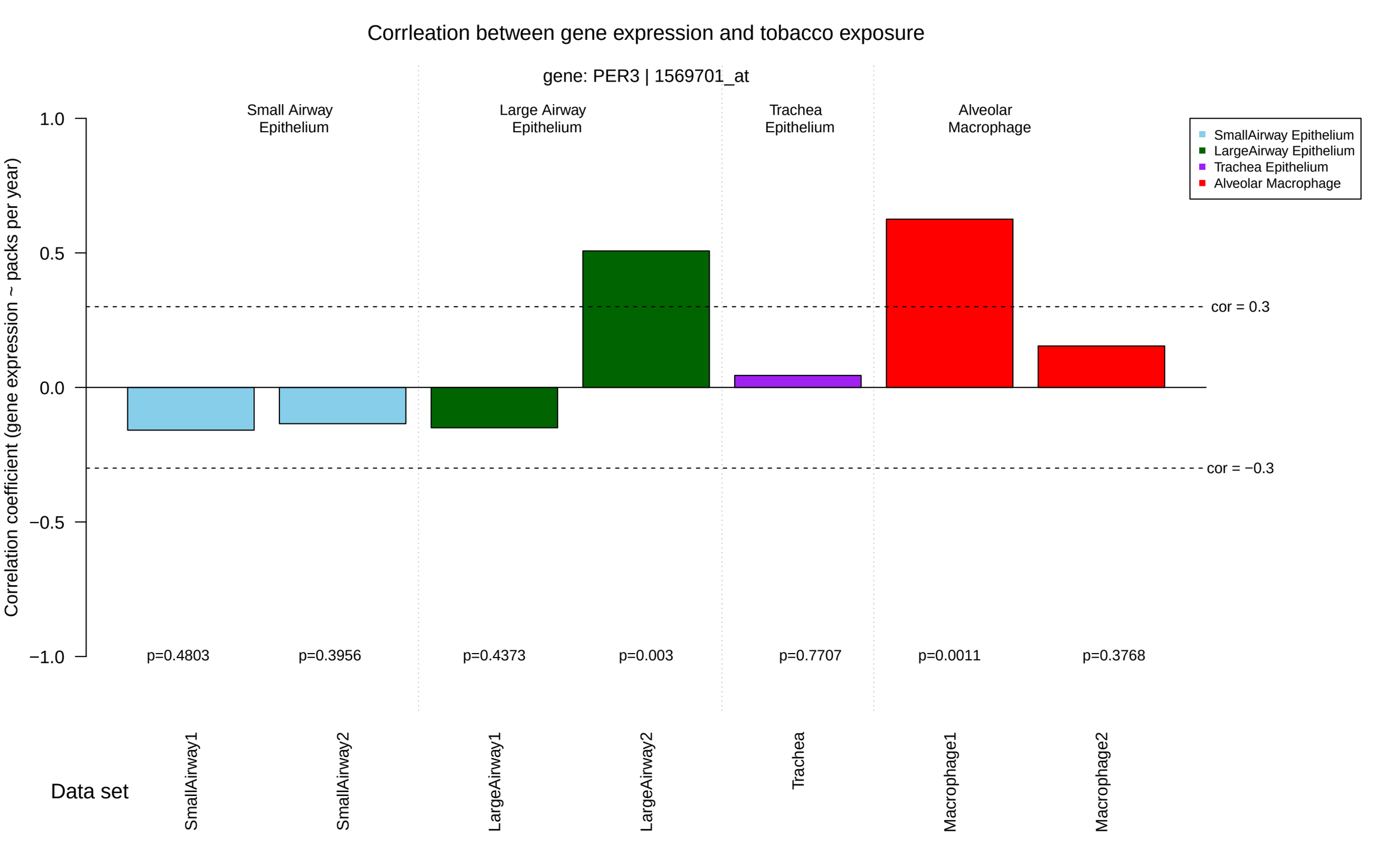

set: LargeAirway1 (p=0.2151)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0422)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.423)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=6e−04)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0217)
PER3 | 1569701_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER3 | 1569701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4803
p=0.3956
p=0.4373
p=0.003
p=0.7707
p=0.0011
p=0.3768
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2